FINAL BUSINESS REPORT

Hotel Revenue & Booking Performance Analytics

An end-to-end analytics project using Excel, MySQL, Power Query, Power BI, and DAX

87,395	27.49%	108.55	3.63 nights
Bookings analyzed	Cancellation rate	Valid-only average ADR	Average stay

Project purpose

Analyze hotel booking demand, cancellation behavior, customer and distribution patterns, and ADR to identify commercially relevant findings and areas for further investigation.

Dataset Hotel Booking Demand Dataset (Kaggle) **Grain** One booking per row **Hotels** City Hotel and Resort Hotel

Prepared as a portfolio project. Figures are based on the cleaned 87,395-row analytical dataset and the validation results recorded during the project.

1. Executive Summary

This project analyzes 87,395 hotel booking records for City Hotel and Resort Hotel. The workflow included data preparation in Excel/Power Query, database loading and analysis in MySQL, and KPI modeling and visualization in Power BI. Core booking counts, cancellation metrics, average stay, guest counts, and ADR calculations were cross-checked across platforms.

The main business pattern is a trade-off between volume and cancellation exposure: City Hotel has more bookings and a higher average ADR, but also a higher cancellation rate than Resort Hotel. Cancellation risk is also associated with longer booking lead times and varies substantially by distribution channel and guest type.

Headline findings

- **Booking volume:** 87,395 bookings; 24,025 canceled bookings; overall cancellation rate 27.49%.
- **Hotel comparison:** City Hotel recorded 53,428 bookings and a 30.04% cancellation rate; Resort Hotel recorded 33,967 bookings and a 23.48% cancellation rate.
- **Lead time:** cancellation rate ranged from 8.43% for 0–7 day lead times to 39.74% for lead times of 181+ days.
- **Distribution:** Online TA accounted for 51,618 bookings with a 35.35% cancellation rate.
- **Guest type:** repeated guests had a 7.64% cancellation rate compared with 28.30% for new guests.
- **Seasonality:** May 2017 had the highest observed monthly booking count (4,575) in the analyzed period.

Recommended management focus

Prioritize cancellation analysis by channel, lead time, and hotel; investigate the unusually high observed cancellation rate for Non-Refund bookings; and use the observed seasonal pattern to inform demand planning. These are investigation and planning recommendations, not guaranteed causal solutions.

2. Business Problem & Objectives

The business needs a consistent view of booking performance across two hotel types, including demand patterns, cancellation exposure, pricing indicators, customer behavior, and distribution-channel performance.

Objectives

- Measure booking volume and cancellation rate overall and by hotel.
- Identify periods with stronger observed booking demand.
- Compare cancellation rates across lead-time groups, market segments, distribution channels, deposit types, and guest types.
- Compare average daily rate (ADR), length of stay, and customer/market patterns.
- Provide practical recommendations while distinguishing observed associations from causation.

3. Dataset & Methodology

Dataset overview

Attribute	Description
Source	Hotel Booking Demand Dataset (Kaggle)
Initial size	119,390 rows; 32 columns in the supplied CSV version
Analytical size	87,395 data rows after exact duplicate removal
Grain	One booking/reservation per row
Coverage	Arrival dates from July 2015 through August 2017
Hotel types	City Hotel and Resort Hotel

Workflow and tools

- **Excel / Power Query:** reviewed the dataset, removed exact duplicates, standardized selected missing values, and created analytical fields.
- **MySQL:** created staging and final tables, loaded the cleaned dataset, validated row counts, and performed exploratory/business analysis.
- **Power BI:** connected to the final analytical table, created a date table and relationship, defined DAX measures, and built a three-page dashboard.
- **Cross-platform validation:** compared key KPI values across Excel, SQL, and Power BI.

Data preparation and analytical fields

Exact duplicate rows were removed, reducing the dataset from 119,389 data rows to 87,395 data rows. Selected missing values were handled: Children nulls were converted to zero, and missing Agent and Company values were labeled No Agent and No Company. Derived fields included ArrivalDate, TotalStayNights, TotalGuests, BookingStatus, RoomChanged, GuestType, StayType, LeadTimeGroup, and ADR_Status.

Data-quality caveats

- One negative ADR record and 1,777 zero ADR records were retained and labeled for analysis rather than silently removed.
- Average ADR is reported both for all rows (106.34) and for ADR_Status = Valid only (108.55). The filtered value is used for pricing analysis.
- A room-type mismatch is treated as an observed operational pattern, not automatically as a data error.
- The dataset contains partial years at the start and end of the observation period; full-year year-over-year growth should not be inferred from those totals.

4. KPI Definitions

KPI	Definition used in this report
Total Bookings	Count of rows in the cleaned analytical table.
Canceled Bookings	Count of records where IsCanceled = 1.
Cancellation Rate	Canceled bookings divided by total bookings.
Average Stay	Average of TotalStayNights.
Average Guests	Average of TotalGuests.

KPI	Definition used in this report
Average ADR — all rows	Average ADR across all rows, including zero and negative values.
Average ADR — Valid only	Average ADR where ADR_Status = 'Valid'.
Estimated Stay Value	ADR × TotalStayNights for eligible non-canceled bookings; a proxy, not actual revenue.

5. Key Performance Results

Overall performance

Metric	Result
Total bookings	87,395
Canceled bookings	24,025
Non-canceled bookings (IsCanceled = 0)	63,370
Cancellation rate	27.49%
Average stay	3.63 nights
Average guests	2.03
Average ADR — all rows	106.34
Average ADR — Valid only	108.55

Hotel comparison

Metric	City Hotel	Resort Hotel
Bookings	53,428	33,967
Average ADR (reported hotel-level result)	110.99	99.03
Average stay	3.15 nights	4.39 nights
Cancellation rate	30.04%	23.48%

Demand and seasonality

The highest observed monthly booking count was 4,575 in May 2017. August had the largest aggregate monthly booking count across the available records (11,257) and a reported average ADR of 150.88. Because the dataset covers partial years at its boundaries, these findings describe the observed period and should not be interpreted as complete annual comparisons.

Cancellation patterns

Group	Bookings	Canceled	Cancellation rate
Lead time 0–7 days	18,304	1,543	8.43%
Lead time 8–30 days	16,340	4,145	25.37%
Lead time 31–90 days	22,744	7,280	32.01%
Lead time 91–180 days	18,243	6,382	34.98%
Lead time 181+ days	11,764	4,675	39.74%
Online TA	51,618	18,245	35.35%
New Guest	83,980	23,764	28.30%
Repeated Guest	3,415	261	7.64%
Non-Refund deposit type	1,038	983	94.70%

The Non-Refund result is unusually high and should be investigated. Its sample is smaller than the No Deposit group, and the observed rate alone does not establish why those bookings were canceled.

6. Business Insights

Insight 1 — Longer lead times are associated with higher cancellation rates

Cancellation rates rise across the lead-time groups, from 8.43% at 0–7 days to 39.74% at 181+ days. This is a strong observed association and supports monitoring advance bookings as a cancellation-risk cohort. It does not prove that booking early causes cancellation.

Insight 2 — City Hotel has higher volume but greater cancellation exposure

City Hotel represents the larger share of bookings and has a higher average ADR, but its cancellation rate is 6.56 percentage points above Resort Hotel. Resort Hotel has a longer average stay. These differences suggest that operational and commercial analysis should be segmented by hotel rather than relying only on portfolio-wide averages.

Insight 3 — Online TA is a high-impact area for cancellation analysis

Online TA contributes 51,618 bookings and has a 35.35% cancellation rate, above the overall rate. Given its volume, deeper segmentation of this channel may help identify where cancellation-management efforts should be focused.

Insight 4 — Repeated guests show lower observed cancellation rates

Repeated guests have a 7.64% cancellation rate versus 28.30% for new guests. This pattern supports evaluating retention opportunities, while recognizing that customer type may be associated with other differences not tested here.

Insight 5 — Observed demand is seasonal

Late spring and summer months show strong observed booking volumes. The pattern can inform initial staffing and pricing planning, but should be tested against complete years and, where available, capacity and occupancy data.

Insight 6 — Non-Refund cancellation results require data validation

The observed 94.70% cancellation rate for Non-Refund bookings is unusual enough to merit a focused audit of category definitions, booking status, hotel, channel, and date. Do not change deposit policy solely on this result.

7. Recommendations

Priority	Recommended action	Why
1	Segment Online TA cancellations by hotel, lead-time group, deposit type, and arrival period.	High booking volume and above-average cancellation rate.
2	Investigate the Non-Refund cancellation pattern and validate field meaning and status consistency.	Unusually high observed rate warrants verification.
3	Monitor advance bookings with longer lead times and test confirmation/reminder approaches.	Cancellation rates increase across lead-time groups.
4	Build seasonal planning around observed demand peaks, then validate with full-year data and capacity metrics.	Demand is not evenly distributed across months.
5	Evaluate customer-retention initiatives using repeat-booking, cancellation, and value measures.	Repeated guests show lower observed cancellation rates.

Measurement principle

Treat recommendations as hypotheses to test. Establish a baseline, implement a controlled or phased change where practical, and measure subsequent cancellation, booking, and value outcomes before concluding that an intervention worked.

8. Limitations & Interpretation

- **Partial-year coverage:** the observed period begins in July 2015 and ends in August 2017. Annual totals are not directly comparable as full-year growth rates.
- **ADR is not revenue:** ADR is a daily rate measure. ADR × stay nights is only an estimated stay-value proxy and should not be reported as actual revenue.
- **Outliers and invalid values:** the dataset contains one negative ADR and zero-ADR records. Valid-only ADR is used for pricing analysis, but the original records remain part of the documented data-quality profile.
- **Association is not causation:** the analysis is descriptive and does not establish that lead time, channel, customer type, or deposit type causes cancellation.
- **Operational context:** occupancy, available room capacity, realized revenue, acquisition costs, and reasons for cancellation are not confirmed in the analyzed fields. Recommendations should be refined if those data become available.

9. Cross-Platform Validation

The following core KPIs were reported consistently across Excel, MySQL, and Power BI: total bookings (87,395), canceled bookings (24,025), non-canceled bookings (63,370), cancellation rate (27.49%), average stay (3.63 nights), and average guests (2.03). ADR was reconciled by calculation scope: 106.34 across all rows and 108.55 for records marked Valid. Room-changed bookings totaled 13,109 in SQL and Power BI; Excel should be confirmed against the same RoomChanged definition if not already checked.

10. Conclusion

The analysis establishes a consistent baseline for hotel booking performance and identifies clear areas for follow-up: cancellation risk among long-lead-time bookings, the high-volume Online TA channel, differences between City and Resort Hotel, and an unusual Non-Refund cancellation pattern. The strongest next step is not to assume causes, but to validate these patterns with more detailed operational data and test targeted improvements.

Tools used

Microsoft Excel • Power Query • MySQL • MySQL Workbench • Power BI • DAX

Dataset reference: Hotel Booking Demand Dataset, Kaggle. The report summarizes results supplied and validated during this project workflow; figures should be rechecked against the final dashboard and SQL scripts before external publication.